Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: a client computer attempted to send a DNS query to a DNS server to resolve the domain name `yummyrecipesforme.com` into an IP address. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable" at timestamp 13:24:32.192571, indicating that the packet could not be delivered The port noted in the error message is used for DNS (Domain Name System) services, which normally listen for incoming connection queries on Port 53 The most likely issue is: the DNS service software on the server is completely turned off, crashed, or blocked by a firewall, preventing the website name from being translated into an IP address

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: 13:24:32 (or 1:24 PM). Explain how the IT team became aware of the incident: Several customers reported that they could not access the company website Explain the actions taken by the IT department to investigate the incident: A cybersecurity analyst attempted to visit the website, confirmed the error, and then used the network analyzer tool `tcpdump` to capture data packets while attempting to load the webpage again Note key findings of the IT department's investigation: The investigation found that the browser sent a UDP request to the DNS server (`203.0.113.2`), but received an ICMP error packet back stating that UDP port 53 was unreachable. This process failed three times consecutively, showing that the DNS service itself is down or crashed

Note a likely cause of the incident: the DNS service software on the server is completely turned off, crashed, or blocked by a firewall, preventing the website name from being translated into an IP address